

Trainings ▾

Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. Refer to the original equipment manufacturer (OEM) service manual.
- Remove rocker lever cover. Refer to Procedure 003-011 in Section 3. (/qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html)



with Electronically Actuated Injector

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⚠ WARNING ⚠

The pressure of the fuel in the line is sufficient to penetrate the skin and cause serious personal injury. Wear gloves and protective clothing.

⚠ CAUTION ⚠

Use caution when disconnecting or removing fuel lines, replacing filters, and priming the fuel system so that fuel is not spilled or drained into the bilge area. Do not drop or throw

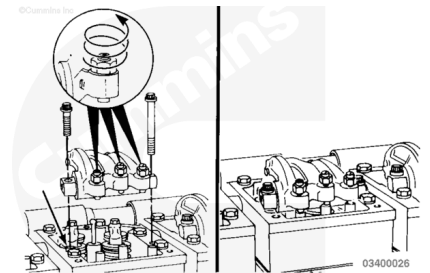
filter elements into the bilge area. The fuel and fuel filter must be disposed of in accordance with local environmental regulations.

- Disconnect the batteries. Refer to the OEM service manual.
- Remove the injector fuel supply lines. Refer to Procedure 006-051 in section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-051-tr.html>)
- Remove rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html>)

Remove with Mechanically Actuated Injector

Do **not** allow the rocker levers to fall off the shaft during removal.

Remove the two mounting capscrews and the rocker levers.

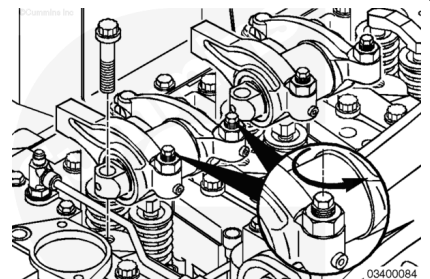


with Electronically Actuated Injector

Loosen the rocker lever adjusting screws.

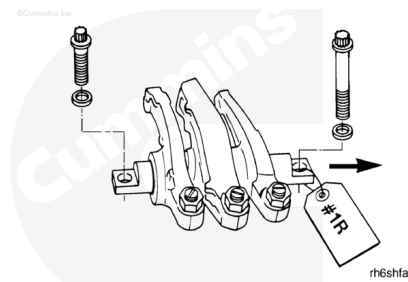
Loosen the rocker lever shaft mounting capscrews and remove the rocker lever assembly.

Do **not** allow the rocker levers to fall off the shaft during removal.

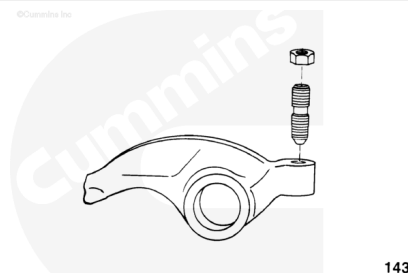


Disassemble with Mechanically Actuated Injector

Remove the capscrews and plain washers. Pull the shaft out. Tag the parts to aid the future assembly procedure.



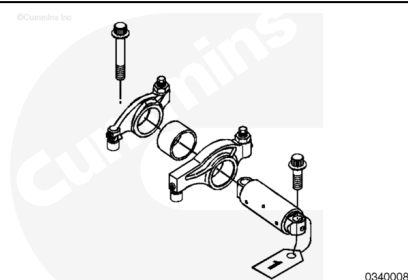
Remove the lock nut and adjusting screw.



with Electronically Actuated Injector

Remove the capscrews.

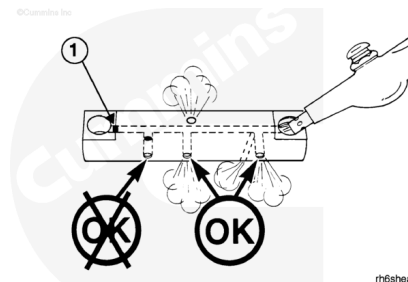
Pull the shaft out of the rocker levers.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the risk of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠

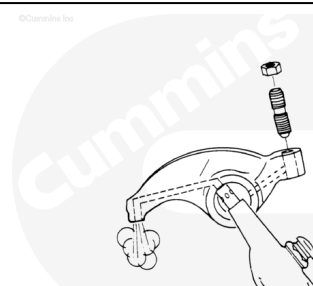
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

Use solvent. Clean the shaft. Use compressed air to blow through the oil drillings, to be sure that the five oil holes are **not** blocked.

The plug (1) in the main oil rifle of the rocker shaft **must** be installed in the ring dowel end of the shaft (exhaust lever end).

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

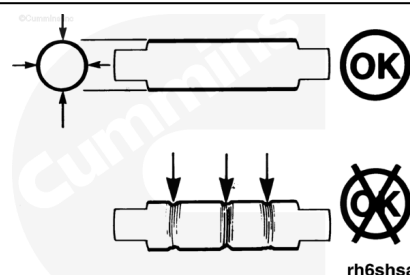
Use solvent. Clean the parts.

Use compressed air to blow through the oil drillings to be sure that they are **not** blocked.

Check the shaft for damage or roughness.

Measure the outside diameter.

Shaft Outside Diameter		
mm		in
34.823	MIN	1.3710
34.862	MAX	1.3725

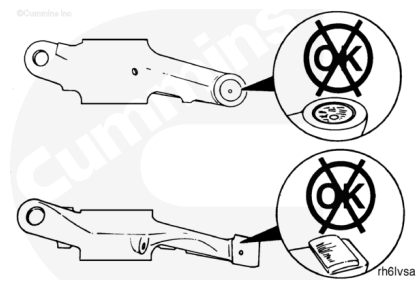


Check the socket of the injector lever for wear. If the socket shows any signs of scratching or dullness, it **must** be replaced. Polishing is expected.

Check the valve rocker lever pads for wear.

If the pad has been worn flat at the point it contacts the crosshead, the lever **must** be replaced. Some polishing of the lever pad is normal.

Note : Engines with electronically actuated injectors do not have injector levers.



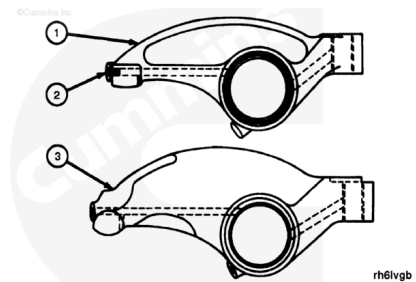
Inspect the rocker levers for loose rivets.

The intake rocker lever (1) **must** have a rivet (2) installed in the oil drilling.

The exhaust rocker lever (3) **must not** have a rivet in the oil drilling.

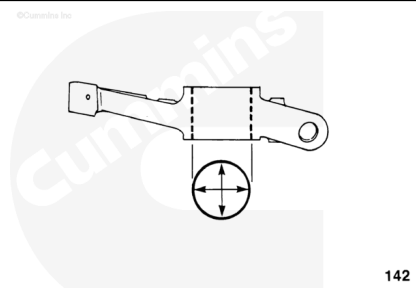
Note : K2000E, K1800E and K1500E engines **must** have a rivet in the intake and exhaust levers.

Note : All engines with electronically actuated injectors **must** have a rivet in the intake and exhaust levers.



Measure the inside diameter.

Bushing Bore Inside Diameter		
mm		in
34.887	MIN	1.3735
34.991	MAX	1.3776

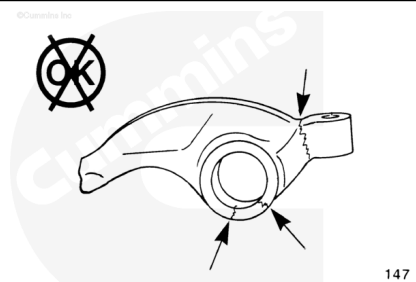


Note : This specification is the same for all rocker levers.

If the bushing bore inside diameter is **not** within specifications, the bushing **must** be replaced.

Use crack detection kit, Part Number 3375432, or magnetic particle method to inspect the rocker levers. If using the magnetic particle method, reference the magnetic crack inspection in this procedure.

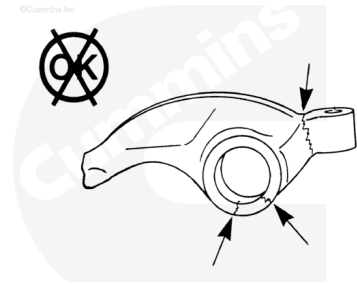
The lever **must** be replaced if there is a crack in any location.



Magnetic Crack Inspect

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



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The rocker lever **must** be disassembled to perform this inspection.

Use the residual method and apply coil shot amperage.

Ampere turns is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.

Amperage (Ampere Turns)

1200-VDC or rectified VAC

2000-VDC or rectified VAC

The lever **must** be replaced if there is a crack in any location.

Demagnetize the lever.

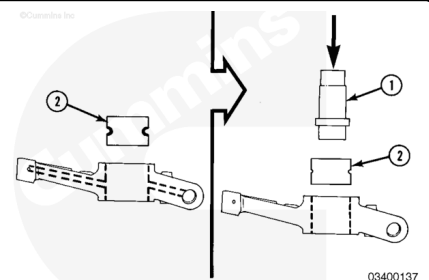
Clean the rocker lever with solvent.

Assemble

with Mechanically Actuated Injector

⚠ CAUTION ⚠

The bushing must be installed and aligned in the bushing bore correctly. A loss of lubricating oil to the shaft will occur if the bushings are incorrectly installed.



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⚠ CAUTION ⚠

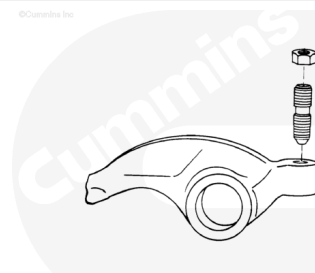
The plug in the end of the shaft must be on the same end as the exhaust rocker lever. Damage to the rocker assembly will occur if the plug is not installed or installed in the wrong location.

Align the slot in the bushing to point toward the nose of the rocker lever while aligning the oil holes in the bushing with the oil passages in the rocker lever.

Use an arbor press and a rocker lever mandrel (1), Part Number 3162458, for mechanically actuated injectors, to install the bushing.

Install the adjusting screw and the lock nut in each lever.

Do **not** completely tighten the lock nut until the assembly is installed on the engine.



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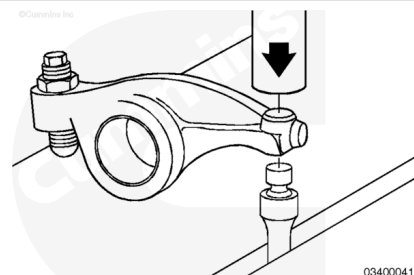
⚠ CAUTION ⚠

Do not overstress the press fit. Damage to the rocker lever can result.

Use an old push tube socket end to support and align the ball for proper seating into the injector rocker lever.

Use a press to seat the new ball into the cavity in the rocker lever.

The lip on the ball **must** be flush with the surface on the rocker lever.



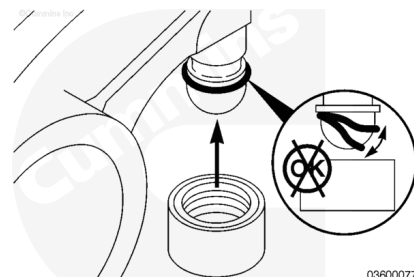
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⚠ CAUTION ⚠

Make sure the o-ring is positioned on the ball properly. If the o-ring is installed between the ball and socket the o-ring will eventually fail, resulting in excessive valve lash.

The o-ring is **only** used to retain the socket to the rocker lever during assembly.

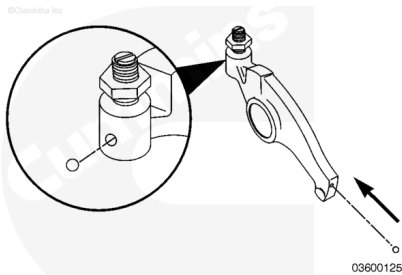
Install the new socket on the ball.



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If removed, install drive plugs into oil drillings in the rocker levers.

Contact an Authorized Cummins® Repair Location for the proper sealing mechanism part number.



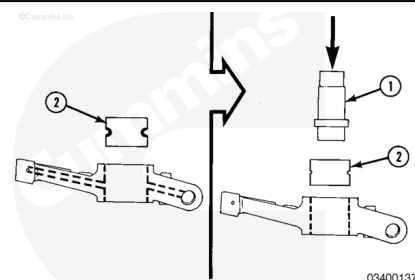
with Electronically Actuated Injector

⚠ CAUTION ⚠

The bushing must be installed and aligned in the bushing bore correctly. A loss of lubricating oil to the shaft will occur if the bushings are incorrectly installed.

⚠ CAUTION ⚠

The plug in the end of the shaft must be on the same end as the exhaust rocker lever. Damage to the rocker assembly will occur if the plug is not installed or installed in the wrong location.

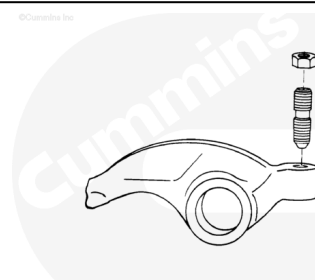


Align the slot in the bushing to point toward the nose of the rocker lever while aligning the oil holes in the bushing with the oil passages in the rocker lever.

Use an arbor press and a rocker lever mandrel (1), Part Number ST-1284, or equivalent, for electrically actuated injectors, to install the bushing.

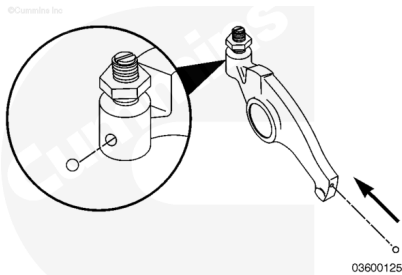
Install the adjusting screw and the lock nut in each lever.

Do **not** completely tighten the lock nut until the assembly is installed on the engine.



If removed, install drive plugs into oil drillings in the rocker levers.

Contact a Cummins® Authorized Repair Location for the proper sealing mechanism part number.



Install

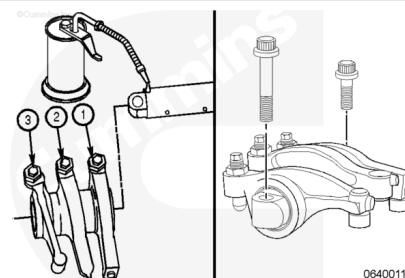
with Mechanically Actuated Injector

⚠ CAUTION ⚠

The oil drillings in the shaft must be in alignment with the oil drillings in the rocker levers.

⚠ CAUTION ⚠

The plug in the end of the shaft must be on the same end as the exhaust rocker lever. Damage to the rocker assembly will occur if the plug is not installed or installed in the wrong location.



Use clean oil to lubricate the shaft.

Slide the rocker levers on the shaft in the sequence shown.

- Exhaust lever (1)
- Injector lever (2)
- Intake lever (3)

Lubricate the threads, rocker sockets, and crosshead pads with clean engine oil.

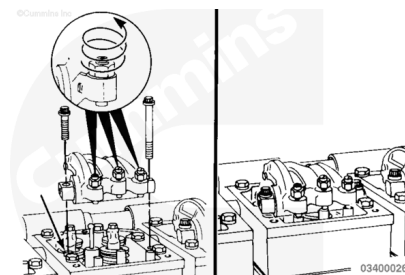
Install the washers and capscrews.

Make sure the ring dowel that aligns the rocker shaft is installed in the housing.

Make sure the adjusting screws are loose.

Position the rocker lever assembly on the housing. Install the capscrews and finger-tighten two or three revolutions.

Align the push rod sockets with the adjusting screws.



Tighten the capscrews.

Torque Value: 95 n•m [70 ft-lb]

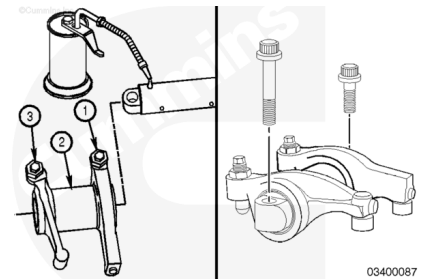
with Electronically Actuated Injector

⚠ CAUTION ⚠

The oil drillings in the shaft must be in alignment with the oil drillings in the rocker levers.

⚠ CAUTION ⚠

The plug in the end of the shaft must be on the same end as the exhaust rocker lever. Damage to the rocker assembly will occur if the plug is not installed or installed in the wrong location.



Use clean oil to lubricate the shaft.

Install the exhaust lever (1), spacer (2), and intake lever (3) onto the shaft. The oil drillings in the shaft **must** align with the drillings in the levers.

Note : The plug in the end of the shaft must be on the same end as the exhaust rocker lever.

Lubricate the threads, rocker sockets, and crosshead pads with clean engine oil.

Install the washers and capscrews.

Make sure the ring dowel that aligns the rocker shaft is installed in the housing.

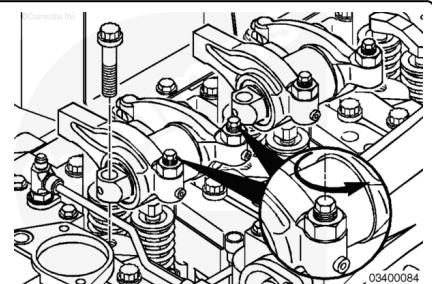
Make sure the adjusting screws are loose.

Position the rocker lever assembly on the housing. Install the capscrews and finger-tighten two or three revolutions.

Align the push rod sockets with the adjusting screws.

Tighten the capscrews.

Torque Value: 95 n•m [70 ft-lb]



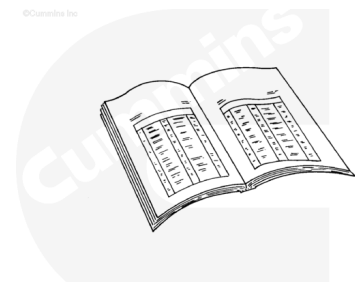
Finishing Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

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ck800wa

⚠ WARNING ⚠

The pressure of the fuel in the line is sufficient to penetrate the skin and cause serious personal injury. Wear gloves and protective clothing.

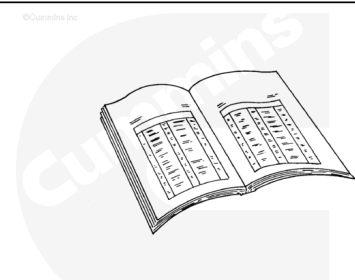
- Adjust the valves and injectors. Refer to Procedure 003-006 in section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-006-tr.html>)
- Install the rocker lever covers. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html>)
- Connect the batteries. Refer to the OEM service manual.
- Operate the engine and check for leaks.

with Electronically Actuated Injector

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ck800wa

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- Adjust the overhead. Refer to Procedure 003-006 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-006-tr.html>)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/28/28-003-011-tr.html>)
- Install the injector supply lines. Refer to Procedure 006-051 in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-051-tr.html>)
- Connect the batteries. Refer to the OEM service manual.
- Operate the engine and check for leaks.